Superhero Directory & 30-Day Schedule — Solution

```
// 1️⃣ Data class
data class Superhero(
    val name: String,
    val alias: String,
    val power: String,
    val favorite: Boolean = false,
    val accessibleDescription: String
)

// 2️⃣ Format hero (pure function)
fun formatHero(hero: Superhero): String {
    val favTag = if (hero.favorite) " [Fav]" else ""
    return "${hero.name} (${hero.alias}) - ${hero.power}$favTag (A11y: ${hero.accessibleDescription})"
}

// 3️⃣ Filter by power (case-insensitive contains)
fun filterByPower(heroes: List<Superhero>, powerQuery: String): List<Superhero> {
    return heroes.filter {
        it.power.lowercase().contains(powerQuery.lowercase())
    }
}

// 4️⃣ Get favorites
fun getFavorites(heroes: List<Superhero>): List<Superhero> {
    return heroes.filter { it.favorite }
}

// 5️⃣ Generate schedule
fun generateSchedule(heroes: List<Superhero>, days: Int): List<String> {
```

```
    val favorites = getFavorites(heroes)
    val sourceList = if (favorites.isNotEmpty()) favorites else heroes

    val schedule = mutableListOf<String>()

    for (day in 1..days) {
        val hero = sourceList[(day - 1) % sourceList.size]
        schedule.add("Day $day: ${hero.name} (${hero.alias})")
    }

    return schedule
}

// 6️⃣ Main
fun main() {

    val heroes = listOf(
        Superhero(
            "Clark Kent",
            "Superman",
            "Flight",
            true,
            "Clark Kent, alias Superman, power Flight"
        ),
        Superhero(
            "Diana Prince",
            "Wonder Woman",
            "Strength",
            true,
            "Diana Prince, alias Wonder Woman, power Strength"
        ),
        Superhero(
```

```
        "Bruce Wayne",
        "Batman",
        "Intelligence",
        false,
        "Bruce Wayne, alias Batman, power Intelligence"
    ),
    Superhero(
        "Barry Allen",
        "Flash",
        "Speed",
        false,
        "Barry Allen, alias Flash, power Speed"
    ),
    Superhero(
        "Arthur Curry",
        "Aquaman",
        "Water Control",
        false,
        "Arthur Curry, alias Aquaman, power Water Control"
    ),
    Superhero(
        "Hal Jordan",
        "Green Lantern",
        "Energy Constructs",
        false,
        "Hal Jordan, alias Green Lantern, power Energy Constructs"
    )
)

// --- Superhero Directory ---
println("--- Superhero Directory ---")
```

```
heroes.forEachIndexed { index, hero ->
    println("${index + 1}. ${formatHero(hero)}")
}

// --- Filter Section ---
val filterQuery = "flight"
val filteredHeroes = filterByPower(h
```